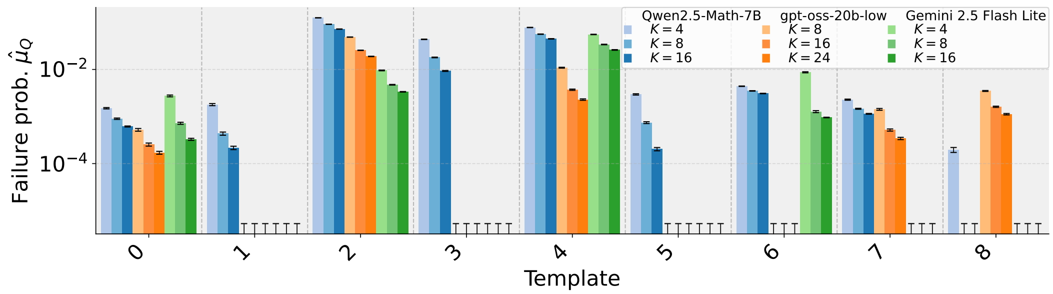

Figure 5: Estimated failure probability $\hat{\mu}_Q$ across templates and models at $\lambda = 0.1$ and evaluation size of $1M$, with error bars denoting the half-width of the confidence interval. For those with zero failures, we report the error bars of exact binomial bounds, i.e., (0, 5.30e-06).

and smaller efficiency gains (Table 2), which is consistent with our finding that stronger failure concentration drives higher importance sampling efficiency.

7 Reliability Evaluation Between Models on GSM

Finally, Figure 5 reports each model's estimated rare failure probability $\hat{\mu}_Q$ across GSM templates, with evaluation size of $M = 1$M, presented as a point estimate with confidence bounds. While all models achieve saturating performance with error rates well below 1%, our evaluation reveals meaningful differences in reliability at finer granularity. Specifically, gpt-oss-20b-low gives a failure rate of just 0.0254% on Template 0 with $K = 16$, which outperforms Qwen2.5-Math-7B-Instruct (0.0614%, $\sim 2.5\times$ higher) and Gemini 2.5 Flash Lite (0.0328%, $\sim 1.3\times$ higher). These results highlight that such saturated benchmarks still carry discriminative signal for reliability, and such evaluation encourages the community to pursue models that are not only more capable but more reliably so. For all template-model pairs that include zero failures, we report the exact binomial bounds, i.e., (0, 0.00053%). The full table is provided in Table 4.

8 Limitations, Future Work, and Conclusion

Limitations. Our method relies on the assumption that LLM errors concentrate on a small subset of the parameter space, which we validate empirically across the models and templates we study. When error patterns are diffuse, e.g., when the model is relatively weak for a task, the CEM may not find a substantially better proposal than uniform sampling. Our method may also be less effective when estimating failure rates without majority voting (K=1), where stochastic generation outputs produce inconsistent failure patterns that are harder to concentrate on.

In addition, due to computational constraints, our experiments focus on relatively small LLMs. Evaluating reliability introduces significant overhead: the required number of inferences grows with the evaluation sample size, the majority-vote ensemble size K, and the number of independent trials for coverage verification, all of which increase as failures become rarer or as the desired confidence level tightens. Extending this framework to larger, more capable LLMs therefore remains an important direction for future work.

Lack of reliability benchmarks. While nearly all existing benchmarks focus on evaluating how LLMs push the frontier in challenging tasks, to our knowledge, no benchmark specifically tests their reliability on already-saturated benchmarks. Rather than revisiting these saturated benchmarks, the community tends to curate novel, more challenging ones, leaving a gap in our understanding of model reliability. In this paper, we take a first step toward addressing this gap by introducing a prototypical framework for evaluating LLM reliability using parameterized GSM templates. However, reliability evaluation must be extended to broader domains, including safety guardrails, code generation for production systems, and medical or legal question answering, where even rare failures can have serious consequences. For instance, GPT-5.2-Thinking achieves a 99.9441% no-violation rate on internal safety benchmarks (OpenAI, 2026), yet the remaining 0.0559% violation rate still corresponds to over 500 failures per million queries, underscoring the importance of rigorous reliability measurement even in near-perfect regimes.

Surgical fixes for systematic failures. Identifying model failures that are consistently triggered by certain input prompts could enable more targeted reliability improvements. Our method directly supports this: the learned proposal distribution Q highlights the failure-inducing parameter values, which practitioners can use as targeted training data or test cases to surgically correct systematic failures via fine-tuning on problematic inputs or adding safeguards rather than retraining broadly. The ultimate goal is to leverage these systematic failure patterns to push model reliability into the five-nines regime, enabling safe deployment in high-stakes applications.

Conclusion. We present an inference-efficient framework for estimating the true error probability of highly accurate LLMs, demonstrated on parameterized GSM problems. LLM failures are strongly concentrated on a small subset of inputs, as evidenced by high total variation distances from the uniform distribution across all models and templates we study. Leveraging this finding, we adopt the CEM to learn a failure-prone sampling distribution, achieving up to 156.22$\times$ inference savings over uniform sampling for equivalent confidence bounds. We call on the community to treat reliability as a first-class evaluation criterion alongside capability; our framework provides a practical foundation for rigorous reliability measurement in high-stakes LLM deployments.

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References

Sandhini Agarwal, Lama Ahmad, Jason Ai, Sam Altman, Andy Applebaum, Edwin Arbus, Rahul K Arora, Yu Bai, Bowen Baker, Haiming Bao, et al. gpt-oss-120b & gpt-oss-20b model card. *arXiv preprint arXiv:2508.10925*, 2025.

Anthropic. Claude opus 4.6 system card, 2026. URL https://anthropic.com/claude-opus-4-6-system-card.

Mansur Arief, Zhiyuan Huang, Guru Koushik Senthil Kumar, Yuanlu Bai, Shengyi He, Wenhao Ding, Henry Lam, and Ding Zhao. Deep probabilistic accelerated evaluation: A robust certifiable rare-event simulation methodology for black-box safety-critical systems. In Arindam Banerjee and Kenji Fukumizu (eds.), *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proceedings of Machine Learning Research*, pp. 595–603. PMLR, 13–15 Apr 2021. URL https://proceedings.mlr.press/v130/arief21a.html.

Yuanlu Bai, Zhiyuan Huang, Henry Lam, and Ding Zhao. Rare-event simulation for neural network and random forest predictors. *ACM Trans. Model. Comput. Simul.*, 32(3), July 2022. ISSN 1049-3301. doi: 10.1145/3519385. URL https://doi.org/10.1145/3519385.

Center for AI Safety, Scale AI, and HLE Contributors Consortium. A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649:1139–1146, 2026. doi: 10.1038/s41586-025-09962-4. URL https://arxiv.org/abs/2501.14249.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. ISSN 00063444, 14643510. URL http://www.jstor.org/stable/2331986.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025. URL https://arxiv.org/abs/2507.06261.

Pieter-Tjerk De Boer, Dirk P Kroese, Shie Mannor, and Reuven Y Rubinstein. A tutorial on the cross-entropy method. *Annals of operations research*, 134(1):19–67, 2005.

B. Efron. Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics*, 7(1):1 – 26, 1979. doi: 10.1214/aos/1176344552. URL https://doi.org/10.1214/aos/1176344552.

ARC Prize Foundation. Arc-agi-3: A new challenge for frontier agentic intelligence, 2026. URL https://arxiv.org/abs/2603.24621.

Google DeepMind. Gemini 3 pro model card, 2025. URL https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-Pro-Model-Card.pdf.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Tim Hesterberg. Weighted average importance sampling and defensive mixture distributions. *Technometrics*, 37(2):185–194, 1995.

Valentin Hofmann, David Heineman, Ian Magnusson, Kyle Lo, Jesse Dodge, Maarten Sap, Pang Wei Koh, Chun Wang, Hannaneh Hajishirzi, and Noah A Smith. Fluid language model benchmarking. *arXiv preprint arXiv:2509.11106*, 2025.

Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=VTF8yNQM66.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 34303–34326. PMLR, 21–27 Jul 2024. URL https://proceedings.mlr.press/v235/maia-polo24a.html.

Nestor Maslej, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Njenga Kariuki, Emily Capstick, Anka Reuel, Erik Brynjolfsson, John Etchemendy, Katrina Ligett, Terah Lyons, James Manyika, Juan Carlos Niebles, Yoav Shoham, Russell Wald, Toby Walsh, Armin Hamrah, Lapo Santarlasci, Julia Betts Lotufo, Alexandra Rome, Andrew Shi, and Sukrut Oak. Artificial intelligence index report 2025, 2025. URL https://arxiv.org/abs/2504.07139.

Mike A. Merrill, Alexander G. Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Yeon Shin, et al. Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces, 2026. URL https://arxiv.org/abs/2601.11868.

METR. Task-completion time horizons of frontier ai models. https://metr.org/time-horizons/, 03 2026.